

Solid non-metals

Unlike metals, most non-metals do not conduct heat or electricity, and are known as insulators. They have other properties that are the opposite of those of metals, too, such as lower melting and boiling points.

On the right side of the periodic table are the elements that are described as non-metals. These include the halogens and the noble gases (see pp.40–41). There is also a set known as “other non-metals,” which contains the elements carbon, sulfur, phosphorus, and selenium, all solids at room temperature. All of these exist in different forms, or allotropes. The “other non-metals” set of elements also includes a few gases (see pp.38–39).

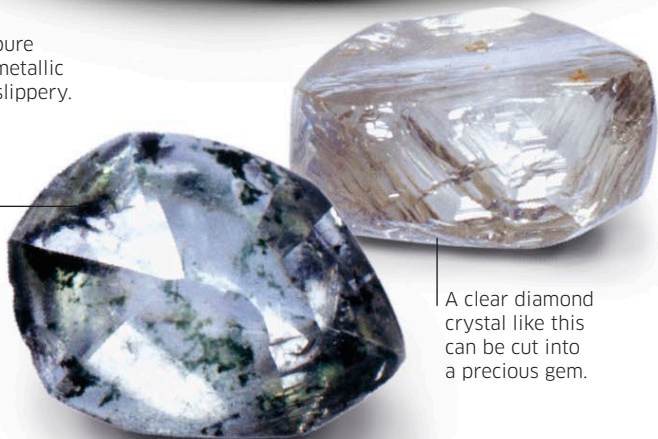


Raw graphite

The surface of pure graphite looks metallic but is soft and slippery.

Raw diamond

Formed deep underground, raw diamonds are found in igneous (volcanic) rocks.



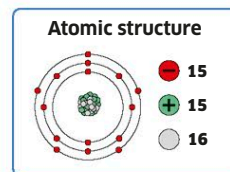
A clear diamond crystal like this can be cut into a precious gem.

PHOSPHORUS

Phosphorus

Discovered: 1669

As a German alchemist boiled urine to produce the mythical philosopher's stone, he discovered a glowing, and very reactive, material instead. He named it phosphorus. It has a number of forms. The two most common are known as red phosphorus and white phosphorus.



Red phosphorus

More stable than white phosphorus, this form is used in safety matches and fireworks.



White phosphorus

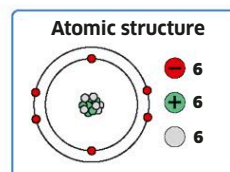
White phosphorus needs to be stored in water because it bursts into flames when in contact with air. It can cause terrible burns.

CARBON

Carbonium

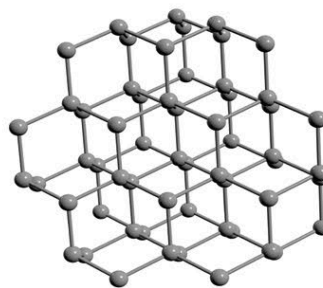
Discovered: Prehistoric times

Carbon is at the center of all life. This element forms the backbone of almost all the most important biological molecules. DNA, amino acids, proteins, fats, and sugars all contain multiple joined carbon atoms, bonded with other atoms, to form the molecules that make living organisms work. Carbon is in our bodies, in our food, in plants, and in most fuels we use for heating and transportation. It appears as crystal-clear diamond as well as soft graphite.



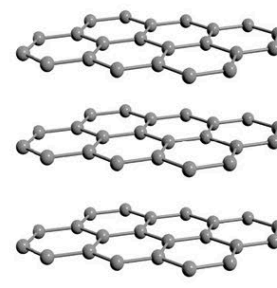
Carbon allotropes

Allotropes are different forms of the same element. Carbon has three main allotropes: diamond, graphite, and buckminsterfullerene. It is the way the carbon atoms are arranged and bonded that determines which allotropes exist, and what their chemical and physical properties are.



Diamond

Diamond, an extremely hard allotrope of carbon, has its atoms arranged in a three-dimensional, rigid structure, with very strong bonds holding all of the atoms together.



Graphite

The “lead” in pencils is actually clay mixed with graphite, an allotrope in which the atoms bond in layers of hexagons. These can slide over each other, making it soft and greasy.